THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{1}{4}$	Work the bracket first.
2	**(a)** 135	Find the number first.
3	**(c)** 30	Time for the whole = time for the part ÷ fraction.
4	**(a)** $\frac{1}{3}$	Unpainted = (unpainted before) × (what is not painted of it).
5	**(b)** greater than 1	Reason about the size of products and quotients.
6	**(a)** 2	Multiply the fraction by the mixed number.
7	**(b)** 1	Find x first.
8	**(d)** $\frac{1}{3}$	Work the bracket first.
9	**(a)** Both A and R are true and R is the correct explanation of A.	6/12 = 1/2.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	(3/4) x 2 = 3/2 and (1/2)(3/4) = 3/8. Both true but separate facts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $= 1\frac{1}{6}$; Difference $= \frac{1}{2}$ (1); Quotient $= 2\frac{1}{3}$; Final $= 1\frac{1}{6}$ (1).	1+1
12	Left $= 1 - \frac{2}{7} = \frac{5}{7}$ of the number; Number $= 35 \div \frac{5}{7} = 49$ (1); Removed $= 14$ (1).	1+1
13	Mean $= \frac{1}{2}\left(\frac{1}{3} + \frac{1}{2}\right)$; $= \frac{5}{12}$ (1); Distance from $\frac{1}{3}$: $\frac{1}{12}$; Distance from $\frac{1}{2}$: $\frac{1}{12}$ (equal) (1).	1+1
14	Sum $= 1\frac{1}{4}$; Difference $= \frac{1}{2}$ (1); Quotient $= 2\frac{1}{2}$; Final $= 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ravi's one day work $= \frac{1}{6}$, Sita's $= \frac{1}{3}$; Together in one day $= \frac{1}{2}$ (1); Days $= 1 \div \frac{1}{2}$ (1); $= 2$ days (1).	1+1+1
16	First distance $= 9$ km; Second distance $= 15$ km (1); Total $= 24$ km in 6 hours (1); Average speed $= 4$ km/h (1).	1+1+1
17	Pieces $= 20 \div \frac{5}{3} = 12$; Full pieces $= 12$ (1); Used $= 20$ m; Left $= 0$ m (1); Final statement (1).	1+1+1
	OR: Ravi's one day work $= \frac{1}{10}$, Sita's $= \frac{1}{15}$; Together in one day $= \frac{1}{6}$ (1); Days $= 1 \div \frac{1}{6}$ (1); $= 6$ days (1).	1+1+1
18	First distance $= 42$ km; Second distance $= 17\frac{1}{2}$ km (1); Total $= 59\frac{1}{2}$ km in 6 hours (1); Average speed $= 9\frac{11}{12}$ km/h (1).	1+1+1
	OR: Pieces $= 25 \div \frac{9}{4} = 11\frac{1}{9}$; Full pieces $= 11$ (1); Used $= 24\frac{3}{4}$ m; Left $= \frac{1}{4}$ m (1); Final statement (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ kg (1). (ii) $4 \times \frac{3}{4} = 3$ kg (1). (iii) $\frac{3}{4} \times \frac{3}{2} = \frac{9}{8} = 1\frac{1}{8}$ kg (1). (iv) $5 \div \frac{3}{4} = \frac{20}{3} = 6\frac{2}{3}$, so 6 full cakes (1); flour used $= 6 \times \frac{3}{4} = 4\frac{1}{2}$ kg, so $\frac{1}{2}$ kg is left (1).	1+1+1+2
20	Sold $= 144$ acres; remainder $= 216$ (1); Given to son $= 36$ acres; remainder $= 180$ (1); Sugarcane $= \frac{4}{5} \times 180 = 144$ acres (1); Unplanted $= 180 - 144 = 36$ acres (1); As a fraction of the whole land, unplanted $= \frac{1}{10}$ (1).	1+1+1+1+1
	OR: A: $\frac{1}{4}$, B: $\frac{1}{12}$ per hour (1); Drain: $\frac{1}{6}$ per hour (emptying) (1); Net per hour $= \frac{1}{6}$ (1); Time $= 1 \div \frac{1}{6} = 6$ hours (2).	1+1+1+2

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
- Inverting the wrong fraction when dividing: only the *divisor* is inverted, not the dividend.
- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change ÷ to ×.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a + sign.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{5}$	Unpainted = (unpainted before) $\times$ (what is not painted of it).
2	**(c)** no positive x	Reason about the size of products and quotients.
3	**(a)** 2	Multiply the fraction by the mixed number.
4	**(a)** 1	Find x first.
5	**(d)** $\frac{1}{3}$	Work the bracket first.
6	**(b)** 135	Find the number first.
7	**(d)** 24	Time for the whole = time for the part $\div$ fraction.
8	**(d)** $\frac{1}{4}$	Unpainted = (unpainted before) $\times$ (what is not painted of it).
9	**(c)** A is true but R is false.	A is true. Dividing by a fraction means multiplying by its reciprocal, so R is false.
10	**(d)** A is false but R is true.	The result is smaller (4 < 8), so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Left $= 1 - \frac{3}{8} = \frac{5}{8}$ of the number; Number $= 45 \div \frac{5}{8} = 72$ (1); Removed $= 27$ (1).	1+1
12	Mean $= \frac{1}{2}\left(\frac{2}{5} + \frac{3}{4}\right)$; $= \frac{23}{40}$ (1); Distance from $\frac{2}{5}$: $\frac{7}{40}$; Distance from $\frac{3}{4}$: $\frac{7}{40}$ (equal) (1).	1+1
13	Sum $= 1$; Difference $= \frac{1}{2}$ (1); Quotient $= 2$; Final $= 1\frac{1}{3}$ (1).	1+1
14	Left $= 1 - \frac{4}{9} = \frac{5}{9}$ of the number; Number $= 60 \div \frac{5}{9} = 108$ (1); Removed $= 48$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ravi's one day work $= \frac{1}{12}$, Sita's $= \frac{1}{4}$; Together in one day $= \frac{1}{3}$ (1); Days $= 1 \div \frac{1}{3}$ (1); $= 3$ days (1).	1+1+1
16	First distance $= 27$ km; Second distance $= 18$ km (1); Total $= 45$ km in 7 hours (1); Average speed $= 6\frac{3}{7}$ km/h (1).	1+1+1
17	Pieces $= 18 \div \frac{7}{4} = 10\frac{2}{7}$; Full pieces $= 10$ (1); Used $= 17\frac{1}{2}$ m; Left $= \frac{1}{2}$ m (1); Final statement (1).	1+1+1
	OR: Ravi's one day work $= \frac{1}{20}$, Sita's $= \frac{1}{30}$; Together in one day $= \frac{1}{12}$ (1); Days $= 1 \div \frac{1}{12}$ (1); $= 12$ days (1).	1+1+1
18	First distance $= 45$ km; Second distance $= 15$ km (1); Total $= 60$ km in 6 hours (1); Average speed $= 10$ km/h (1).	1+1+1
	OR: Pieces $= 30 \div \frac{7}{2} = 8\frac{4}{7}$; Full pieces $= 8$ (1); Used $= 28$ m; Left $= 2$ m (1); Final statement (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{3}{4} \times \frac{2}{5} = \frac{3}{10}$ m² (1). (ii) $2\left(\frac{3}{4} + \frac{2}{5}\right) = 2 \times \frac{23}{20} = \frac{23}{10} = 2\frac{3}{10}$ m (1). (iii) $\frac{1}{3} \times \frac{3}{10} = \frac{1}{10}$ m² (1). (iv) $\frac{3}{2} \times \frac{2}{3} = 1$ m² (1); it is larger by $1 - \frac{3}{10} = \frac{7}{10}$ m² (1).	1+1+1+2
20	Sold $= 30$ acres; remainder $= 150$ (1); Given to son $= 60$ acres; remainder $= 90$ (1); Sugarcane $= \frac{3}{8} \times 90 = 33\frac{3}{4}$ acres (1); Unplanted $= 90 - 33\frac{3}{4} = 56\frac{1}{4}$ acres (1); As a fraction of the whole land, unplanted $= \frac{5}{16}$ (1).	1+1+1+1+1
	OR: A: $\frac{1}{8}$, B: $\frac{1}{24}$ per hour (1); Drain: $\frac{1}{12}$ per hour (emptying) (1); Net per hour $= \frac{1}{12}$ (1); Time $= 1 \div \frac{1}{12} = 12$ hours (2).	1+1+1+2

Mistakes to avoid:

- Multiplying mixed numbers by multiplying the whole parts and the fractions separately; convert to improper fractions first.
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- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change $\div$ to $\times$.
- Believing multiplication always makes a number bigger; multiplying by a proper fraction makes it smaller.
- Forgetting to cancel common factors, or cancelling across a $+$ sign.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $1\frac{3}{5}$	Find x first.
2	**(c)** $\frac{1}{8}$	Work the bracket first.
3	**(a)** 63	Find the number first.
4	**(a)** 24	Time for the whole = time for the part ÷ fraction.
5	**(a)** $\frac{5}{12}$	Unpainted = (unpainted before) × (what is not painted of it).
6	**(a)** $6 \div \frac{2}{3}$	Reason about the size of products and quotients.
7	**(d)** 1	Multiply the fraction by the mixed number.
8	**(d)** $\frac{1}{2}$	Find x first.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R shows the working that gives A.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	No number multiplied by 0 gives 1, so A is true. R is true but unrelated.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mean $= \frac{1}{2}\left(\frac{1}{6}+\frac{5}{6}\right); = \frac{1}{2}$ (1); Distance from $\frac{1}{6}$: $\frac{1}{3}$; Distance from $\frac{5}{6}$: $\frac{1}{3}$ (equal) (1).	1+1
12	Sum $= \frac{5}{6}$; Difference $= \frac{1}{2}$ (1); Quotient $= 1\frac{2}{3}$; Final $= 1\frac{4}{21}$ (1).	1+1
13	Left $= 1 - \frac{5}{11} = \frac{6}{11}$ of the number; Number $= 42 \div \frac{6}{11} = 77$ (1); Removed $= 35$ (1).	1+1
14	Mean $= \frac{1}{2}\left(\frac{3}{8}+\frac{7}{12}\right); = \frac{23}{48}$ (1); Distance from $\frac{3}{8}$: $\frac{5}{48}$; Distance from $\frac{7}{12}$: $\frac{5}{48}$ (equal) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ravi's one day work $= \frac{1}{8}$, Sita's $= \frac{1}{24}$; Together in one day $= \frac{1}{6}$ (1); Days $= 1 \div \frac{1}{6}$ (1); $= 6$ days (1).	1+1+1
16	First distance $= 33$ km; Second distance $= 33$ km (1); Total $= 66$ km in 10 hours (1); Average speed $= 6\frac{3}{5}$ km/h (1).	1+1+1
17	Pieces $= 17 \div \frac{6}{5} = 14\frac{1}{6}$; Full pieces $= 14$ (1); Used $= 16\frac{4}{5}$ m; Left $= \frac{1}{5}$ m (1); Final statement (1).	1+1+1
	OR: Ravi's one day work $= \frac{1}{9}$, Sita's $= \frac{1}{18}$; Together in one day $= \frac{1}{6}$ (1); Days $= 1 \div \frac{1}{6}$ (1); $= 6$ days (1).	1+1+1
18	First distance $= 75$ km; Second distance $= 35$ km (1); Total $= 110$ km in 9 hours (1); Average speed $= 12\frac{2}{9}$ km/h (1).	1+1+1
	OR: Pieces $= 22 \div \frac{8}{3} = 8\frac{1}{4}$; Full pieces $= 8$ (1); Used $= 21\frac{1}{3}$ m; Left $= \frac{2}{3}$ m (1); Final statement (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{7}{2}$ (1). (ii) $\frac{8}{7}$ (1). (iii) $\frac{7}{2} \div \frac{7}{8} = \frac{7}{2} \times \frac{8}{7} = 4$ pieces (1). (iv) $\frac{7}{2} \div \frac{3}{4} = \frac{14}{3} = 4\frac{2}{3}$, so 4 full pieces (1); length used $= 4 \times \frac{3}{4} = 3$ m, so $\frac{1}{2}$ m is left (1).	1+1+1+2
20	Sold $= 120$ acres; remainder $= 180$ (1); Given to son $= 100$ acres; remainder $= 80$ (1); Sugarcane $= \frac{3}{5} \times 80 = 48$ acres (1); Unplanted $= 80 - 48 = 32$ acres (1); As a fraction of the whole land, unplanted $= \frac{8}{75}$ (1).	1+1+1+1+1
	OR: A: $\frac{1}{6}$, B: $\frac{1}{9}$ per hour (1); Drain: $\frac{1}{18}$ per hour (emptying) (1); Net per hour $= \frac{2}{9}$ (1); Time $= 1 \div \frac{2}{9} = 4\frac{1}{2}$ hours (2).	1+1+1+2

Mistakes to avoid:

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- Dividing numerators and denominators directly without inverting, or inverting and forgetting to change ÷ to ×.
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